1.

1.1.

2022 pp. 40 - 41

1.2.

2022 pp. 43, 47 - 54

12 $X\mathrm{cm}^2$ Y g

(1) $Y = aX + b$

(2)

(3)

$$(X, Y) = (11, 23), \quad (23, 49), \quad (5, 10), \quad (50, 115), \quad (32, 64), \quad (21, 45)$$
$$(16, 35), \quad (39, 84), \quad (25, 50), \quad (45, 100), \quad (5, 11), \quad (34, 70)$$

2.

2.1.

2022 2022 2022 p. 65

(1)

(1)

(2) (2) (2)

(1) (2)

(3) 1 6 (2)

1 2 3 4 5 6 (2) $\Omega = \{1, 2, 3, 4, 5, 6\}$ (3)

(2) Ω (1)

(3) (2)

2.2.

2022 2022 2022 pp. 78, 80, 83

(1)

(2)

(3)

(4) 3

2.3.

2022 2024 2026 p. 78

1 X

c

$P(X = x) = c \times x \quad (x = 1, 2, 3, 4, 5, 6)$ $\times$

(1) c

(2) $x = 1, 2, 3, 4, 5, 6$ $P(X = x)$

(3) X $E[X]$

(4) $P(X \geq 4)$

2.4.

2023 pp. 81, 86

$$f(x) = \begin{cases} \dfrac{c}{x^3} (x \geq 10) \\ 0 \quad (x < 10) \end{cases}$$

(1) c

(2) $F(x)$

(3) $P(x \geq a) = \dfrac{1}{2}$ a

2.5.

2024 pp. 81, 86

c $f(x) = \begin{cases} c(1-x)(0 \leq x \leq 1) \\ 0 \qquad\qquad (\text{otherwise}) \end{cases}$

(1) c

(2) $P\left(\dfrac{1}{2} < x \leq \dfrac{2}{3}\right)$

(3) $F(x)$

(4) $P\left(\dfrac{1}{2} \leq x < 3\right)$

2.6.

2022 2023 2024 pp. 134, 151, 180

(1)

(2)

(3)

(4)

2.7.

2023 2023 pp. 134, 151, 180

$X_1, X_2, ..., X_n$ Y 2

(1) $X_1, X_2, ..., X_n$ Y

(2) 2 /

(3)

(4)

(5)

(6) 1

2.8.

2025

(1) X μ σ^2

(2) 200 60 6 42
78

3.

3.1.

2024 p. 107

6 4 1 2

(1) 2

(2) 2

3.2.

2026 p. 107

3 5

(1) 1 1

(a) 2

(b) 2

(c) 2 1

(2) 2

1

(3) (1) (2)

3.3.

2022 pp. 107 - 108

1 2 3 4 2

X_1 X_2 $Y = |X_1 - X_2|$

(1) Y

(2) Y

(3) Y

3.4.

2023 2025 pp. 113 - 114

(X, Y)

$$f(x,y)=\begin{cases}6e^{-2x-3y} & (x\geq 0, y\geq 0)\\ 0 & (\text{otherwise})\end{cases}$$

X Y

3.5.

2026

X Y

$$\int_0^\infty e^{-x}dx=1, \qquad \int_0^\infty xe^{-x}dx=1$$

(1) $f(x,y)=\begin{cases}xe^{-x-y} & (x\geq 0, y\geq 0)\\ 0 & (\text{otherwise})\end{cases}$

(a) $f_X(x)$ $f_Y(y)$

(b) X Y

(2) $f(x,y)=\begin{cases}c(x+y)e^{-x-y} & (x\geq 0, y\geq 0)\\ 0 & (\text{otherwise})\end{cases}$

(a) c

(b) X Y

3.6.

2022 pp. 103 - 104, 113

1 4 2 2 1

$X=0$ $X=1$ 2

$Y=0$ $Y=1$

(1) (X,Y)

(2) X Y

3.7.

2022 pp. 67, 114 - 115

2 A B

3.8.

2022 2023 2026 p. 115

3 $\frac{1}{2}$

(1) 1 2

(2) 1 2 A 2 3
B A B

(3) $P(A|B)$

3.9.

2024 p. 115

2 A 4 B 2
7 C 2 9

(1) A B

(2) A C

3.10.

2023 pp. 72 - 73, 109 - 110, 114 - 115

2 E F $P(E) = a$ $P(F) = b$

(1) $P(E \cup F)$ a b

(2) $P(E \cap F^c | E \cup F)$ a b F^c F

3.11.

2025

1 5
X Y $Z = X - Y$

(1) X Y

(2) Z

(3) Z

(4) Z

3.12.

2025 pp. 118 - 119

(1) 2 A B

(2) 500 1 X X
2 K_1 K_2 $3:1$
K_1, K_2 0.98 0.90
0.01
K_1

3.13.

2022 2023 pp. 118 - 119

3 5 1
2

(1) 1

(2) 2 1

3.14.

2023 pp. 118 - 119

1 10% 3

(1)

(2) 2

3.15.

2022 pp. 118 - 119

5 4 3
1 100 g 45 g 30 g

(1) 1
40 g

(2) 2
100 g 2

(3) 3
150 g 3

3.16.

2022 2024 pp. 118 - 119

A 0.1%
A /
99% 10%

(1)

(2) A 1

3.17.

2023 pp. 118 - 119

99%
99% 0.5%
1

(1)

(2)

(3)

(4)

3.18.

2023 pp. 118 - 119

S S
K

- S K 99% 1%
- S K 99% 1%

(1) K S

(2) (1)

3.19.

2025 pp. 118 - 119

$\boldsymbol{A}$

$\boldsymbol{B}$

(1) $P(A|B) = 0.98$ $P(A|B^c) = 0.005$ $P(B) = 0.001$

(2) $P(A|B^c) = 0.003$ $P(B) = 0.0005$ $P(B|A) \geq 0.142$ $P(A|B)$

3.20.

pp. 118 - 119

500 1 XXX 2

K_1 K_2 $K_1 : K_2 = 3 : 1$

$\boldsymbol{K_1}$ 0.98

$\boldsymbol{K_2}$ 0.90

0.01

(1) K_1

(2) (1)

3.21.

2023 pp. 122 - 124

p $0 < p < 1$ X Y

$$f(k) = \begin{cases} p & (k = 1) \\ 1 - p & (k = -1) \end{cases}$$

$Z = XY$

(1) $E[Z]$

(2) X Z $E[(X - E[X])(Z - E[Z])]$

(3) X Z p p Y Z

(4) (3) p $\Pr[X + Y + Z \leq 2]$

3.22.

2023 2024 pp. 125 - 126

2 X Y

(1) a

(2) X Y $\mathrm{Cov}(X, Y)$

(3) X Y

	$Y = 1$	$Y = 10$	$Y = 100$
$X = 1$	a	a	a
$X = 2$	0	a	a
$X = 3$	a	a	a

3.23. n

2023 p. 127

n 1 n $(n + 1)$

4.

4.1.

2025

2 $B(2, p)$

4.2.

2024 p. 141

1 6

(1) 10 5

(2) 100 25

4.3.

2023 2026 pp. 93 - 95, 141

p n

X 2 $\mathrm{Bin}(n, p)$ $q = \left(\dfrac{X}{n}\right)$

(1) $P(|q - p| \leq 0.02) \geq 0.95$ n

(2) (1) n (2)

(3) p

(3) (1) -

n

(4) (2) (3) n

4.4.

2024 pp. 143 - 145, 153

(1) :

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (-\infty < x < \infty)$$

$y = f(x)$ (x, y)

(2) /

(3)

4.5.

2022 pp. 121 - 122, 143

(1) X Y $N(170, 6^2)$

(a) $W = \dfrac{X+Y}{2}$

(b) X Y W

(2) X Y

$$f(x) = \frac{1}{2\pi} \quad (0 \leq x \leq 2\pi), \qquad g(x) = \frac{1}{12} \quad (-3 \leq x \leq 9)$$

$W = XY$

4.6.

2024 pp. 147 - 151

(1) X $N(\mu, \sigma^2)$

(a) $P(\mu - \sigma \leq X \leq \mu + \sigma)$

(b) $P(\mu - 2\sigma \leq X \leq \mu + 2\sigma)$

(c) $P(\mu - 3\sigma \leq X \leq \mu + 3\sigma)$

(2) (

)

(3)

(a) $z(0.01)$

(b) $\Phi(z_0) = 0.01$ z_0

(c) $z(0.005)$

4.7.

2024 pp. 149 - 151

(1) a Z $N(0, 1)$

$$P(|Z| \geq a) = 2P(Z \geq a)$$

(2) X $N(m, \sigma^2)$

$$P\left(|X - m| \geq \frac{\sigma}{4}\right)$$

4

(3) m σ n

$\overline{X}$

$$P\left(|\overline{X} - m| \geq \frac{\sigma}{4}\right) \leq 0.02$$

n

4.8.

2024 2026 pp. 149 - 150

4

E 350,198 66.31

23.55 X $N(66.31, (23.55)^2)$

(1) $P(X \geq 80)$

(2) (3-a)

(3) 80 95

(4) 20%

4.9.

2025 pp. 149 - 150

4

120

10 100 15

C

(1) 95% C

(2)

4.10.

2022 pp. 153 - 154

(1) 6

(2) 10000

(3) 10000 4900 5100

4.11.

pp. 93 - 95, 153 - 154

200 60 6

X

(1) $42 \leq X \leq 78$

(2)

(3) X $N(60, 6^2)$ $P(42 \leq X \leq 78)$

5.

5.1.

2023 2025 pp. 168 - 169

2

5.2.

2024 pp. 93, 168 - 169

1 ATM ATM 1 40
ATM 1 60
ATM
/

60

75

90

105

120

5.3.

2023 pp. 166 - 169

(1) 3 5

(2) 1 2 1 4

5.4.

2025

4

(1) 1 3
2

(2) X $P(X=3) = 5P(X=5)$

$P(X \leq 3)$

5.5.

2023 pp. 166 - 169

1% 200

(1) X

(2)

(3) 3

(2) (3)

5.6.

2024 pp. 153 - 154, 166 - 169

4

(1) 240 6 38 45

(2) 0.1% 500 3

5.7.

2026

$p = 0.001$ $n = 500$

X

$$e^{-0.5} \approx 0.6065, \qquad \sqrt{0.5} \approx 0.7071$$

(1) (a) X

(b) $E[X]$ $\mathrm{Var}(X)$

(2)

(a)

(b) $P(X \geq 3)$

(3)

(a)

(b) $P(X \geq 3)$

(4)

(a) (2) (3)

(b)

np

5.8.

2022 2023 pp. 139 - 143, 168 - 169

p $0 < p < 1$ $(1-p)$ N $N > 1$

(1) $N = 5$

(2) t $0 \leq t \leq N$

(3) m $m =$

N $0 \leq m \leq N$ m

(4) n $0 \leq n \leq N$ $N - n$ $P(n\&N, p)$

(5) (4) n

(6) n $\lambda = Np$ $N \to \infty$ (4) $P(n\&N, p)$

$P(n; \lambda)$ $\lim_{N\to\infty}\left(1+\frac{x}{N}\right)^N = e^x$

6.

6.1.

2022 p. 174

2

(1)

(2)

6.2.

2025

μ σ^2 n $X_1, X_2, ..., X_n$.

(1) $X_1, X_2, ..., X_n$

(2) .

(3) n $S_n = X_1 + X_2 + \cdots + X_n$

.

(4)

6.3.

2025 p. 179

1180 20

(1) 25 1170

(2) N 0.9 1175

N

6.4.

2025

1.2 kg 0.1 kg

100 122.5 kg

6.5.

2022 pp. 202 - 205

μ σ^2 3 X_1 X_2 X_3

(1) .

(a) $T_1 = X_1 + X_2 - X_3$

(b) $T_2 = \dfrac{X_1 + 2X_2 + X_3}{4}$

(c) $T_3 = \dfrac{X_1 + X_2 + X_3}{3}$

μ

(2) $\{T_1, T_2, T_3\}$

6.6.

2023 2023 2025 pp. 176 - 180, 202 - 205

(1) 0 1.2 3

$\{-1, 0, 1\}$

(a)

(b)

(c)

(2)

(3)

(4) θ $\Theta_n = T(X_1, X_2, ..., X_n)$ 3

(a)

(b)

(c)

(5) 3

6.7.

2022 2024 p. 205

1000 10

$$32, \quad 86, \quad 78, \quad 49, \quad 48, \quad 77, \quad 97, \quad 49, \quad 51, \quad 74$$

1000

6.8.

2025 pp. 179, 216

p n

X

(1) X .

(2) n X

(3) p_0 95%

n

(4) $p = 0.05$ $n = 1900$ $P(76 \leq X \leq 114)$

6.9.

2022 pp. 180 - 182

p p 10,000

3,500 $p = \dfrac{3500}{10000} = 0.35$

(1) p

(2) p 2

6.10.

2025

(1) n A Y
$q = \dfrac{Y}{n}$ A p $100(1-\alpha)\%$

(2) S TV T
140 32 T

(a) p 95%

(b) p 3% 0.95

(c)

6.11.

2024 2024 pp. 223 - 225

(1) $Bin(1, p)$ $X_1, X_2, ..., X_n$

(a)

(b)

(2) n A Y

(a) $q = \dfrac{Y}{n}$ A p $100(1-\alpha)\%$

(b)

(3) S TV T
140 32 T

(a) p 95%

(b) p 3% 0.95

(c)

6.12.

2024 pp. 223 - 225

(1) p 0.8 0.9 0.99

$\hat{p}$ p 0.02

(2) p 0.95 $\hat{p}$

p 0.03

6.13.

2025

2 p S $1-p$ F

m S k $B(m;p)$

(1) $B(m;p)$ $f(k)$.

(2) X $B(m;p)$ X .

(3) p $l(p)$.

(4) p .

(5) . .

(a) (4) ?

(b) (4) ?

(c) (4) ?

6.14.

2025

$N(\mu, \sigma^2)$ μ σ^2

(1) μ

(2) σ^2

6.15.

2025

(1)

(2)

/ σ^2 n

N

(3) /

7.

7.3.

2023 pp. 2235 - 236

8000

25 7700

5%

640

7.4.

2023 pp. 238 - 240

T

S 365 6

$$330, \quad 350, \quad 385, \quad 321, \quad 340, \quad 365$$

5%

7.5.

2025 pp. 238 - 240

T

S 360 6

$$350, \quad 355, \quad 385, \quad 390, \quad 370, \quad 365$$

(1) 10%

(2) 5%

7.6.

2023 2024 2024 2025 pp. 235 - 236

(1) μ $N(\mu, 49)$ 49

$H_0 : \mu = 4,$ $H_1 : \mu > 4$

m

(a) 0.05 $m = 6$ 2

(b) 0.01 $m = 7$ 2

(2) $H_1 : \mu = 5$ 0.05

2 0.2 0.1

7.7.

2022 pp. 241 - 242

(1) 2 A B 2

10 12

A 2.04 0.03 B

2.02 0.02 2

5%

2

(2) A B

5%

7.8.

2022 pp. 241 - 242

100 70

68 15

5%

7.9.

2024 pp. 252 - 253

3 1

(1) A 10 A

3 1 A

5%

(2) 10 7 3 1

7.10.

2024 pp. 252 - 253

5 1 5%

7.11.

2023 pp. 252 - 253

10 X p

$H_0 : p = \frac{1}{2}$ H_0

(1) X

(2) X

(a) $H_1 : p < \frac{1}{2}$ 5%

(b) $H_1 : p \neq \frac{1}{2}$ 2%

7.12.

2023 pp. 252 - 253

5

1 p

(1) 5 5 0

(a) $p = \frac{1}{2}$

(b) $p = \frac{2}{3}$

(2) $p \geq \frac{1}{2}$

(a) 3

$p = \frac{1}{2}$

(b) $p = \frac{3}{4}$ 3

7.13. 2

256 116 5%

.

(1) 2

(2) 2 10%

7.14.

2023 2025 pp. 252 - 253

p $0 < p < 1$

$H_0 : p = \frac{1}{2}$ 3 3

3 H_0 H_0

(1) 1 1

(2) 2 2

(3) 2 $\frac{1}{3}$ p

7.15.

2024 pp. 252 - 253

3 A B C A 3 B 2 C 1 1

1

(1) A A $\frac{1}{16}$ B B $\frac{1}{9}$ C C $\frac{1}{25}$

1

(2) 960 640

(1) 5%